B.Sc (Hons) Information Technology
IE3041 - Data Management and Business Intelligence
2025 Jan - June
Assignment

Instructions:

You have **1 hour and 30 minutes** to complete this test and submit your answers.

- **Dataset:** Review the '**Football_Player_Stats_2021_to_2024'** dataset carefully before answering the questions.
- **Answers:** For each question, take a screenshot of your answer in Power BI and insert it into a Word document.
- **File Naming:** Save both your Power BI file and Word document using your **student ID and name** as the file name. Convert the Word document to a **PDF** before submission.
- **Submission:** Ensure that each question includes a corresponding screenshot. If a screenshot is missing, **half marks** will be given for that question.
- **Submission Format:** Submit your work by uploading the Power BI file and the PDF file separately to their respective submission links.
- **Device Restrictions:** Students are **not allowed** to use any mobile phones or other smart devices during the test. You are only permitted to use **Power BI, Word, and the SLIIT Courseweb**.

ANY VIOLATION OF THESE INSTRUCTIONS WILL RESULT IN A **ZERO** FOR THE TEST.

Questions:

Load the data available in 'Football_Player_Stats_2021_to_2024' into Power BI.

1. Add an index column starting from 1 to the dataset and make it the first column.
2. Create a new column named "Player Contribution" which is the sum of "Goals Scored" and "Assists."
3. Add a calculated column to compute the "Goal Scoring Efficiency" by dividing "Goals Scored" by "Matches Played." Round the result to two decimal places.
4. Create a "Player Category" column with categories:

   "Elite" if Goals ≥ 20

   "Pro" if Goals are between 10 and 19

   "Amateur" if Goals are between 1 and 9

   "Rookie" if Goals = 0
5. Create a new column to separate the country code (e.g., "ARG") and the nationality (e.g., "Argentina") from the "Nationality Code" column.
6. Create a measure to calculate the total number of appearances made by players whose position is "Forward" across all seasons in the dataset. (Include the DAX formula for this measure.)
7. Create a bar chart that shows the total goals scored by each team.
8. Add a bubble map to show the count of players in each country.
9. Use a slicer for the "Season" column to allow users to filter data by individual years. Create a line chart that shows the goals scored by each player over the selected seasons.
10. In a chart showing total goals scored by each team, display the average goal level and its value in a different color.

Marking Guide

Criteria	**Marks**
Get data into Power BI	1
Index column	1
Create column – Player Contribution	5
Create column – Goal Scoring Efficiency	5
Create column – Player Category	5
Create column – Nationality	5
Create measure – Total appearances for Forwards	5
Create bar chart – Total goals by team	5
Create bubble map – Count of players per country	6
Create slicer and line chart for Goals scored over seasons	7
Display average goals scored across all teams	5
Total	50